Lab assignment 1

Task 1 *Analysis of spatiotemporal data (eye-tracking data)*

The data file contains data from an eye-tracking session lasting just under 5 minutes. The raw data has been pre-processed to contain just fixations, not every eye sample.

There are 6 fields per sample:

1. A timestamp for the event.
2. A fixation index which is just an order in the file.
3. The event duration in milliseconds.
4. A gaze point index (which is pretty meaningless).
5. X coordinate of gaze point in pixels.
6. Y coordinate of gaze point in pixels.

The objective is to identify regions of interest to the user and observe how t
Questions of interest to be answered are:

- How many regions/areas of interest can be identified?
- How many are heavily used and when?
- Which regions are only used for part of the analysis procedure?
- What are the frequent transitions between the areas of interest?
- How do those transition patterns change over time?

Plot fixations → e.g. scatter plot (transparency), heatmap/kernel density
Extract regions/areas of interest → cluster (density based), tessellate, classify

E.g. Plot frequency of region visits over time, animate fixation plot

Extract flows between regions, regions transitions matrix
Map flows with magnitude

Plot transitions over time, create a space-time cube, animate